

Abbott's Understanding Analysis Exercises

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Exercise 4.3.11 (Contraction Mapping Theorem)

Problem: Let f be a function defined on all of $\mathbb{R}$, and assume that there is a constant c such that $0 < c < 1$ and $|f(x) - f(y)| \leq c|x - y|$ for all $x, y \in \mathbb{R}$.

(a) Show that f is continuous on $\mathbb{R}$.

Proof. Let $\epsilon > 0$ be given. Choose $\delta = \frac{\epsilon}{c}$.

For any $x, y \in \mathbb{R}$ with $|x - y| < \delta$, we have:

$$|f(x) - f(y)| \leq c|x - y| < c\left(\frac{\epsilon}{c}\right) = \epsilon$$

Thus, f is continuous on $\mathbb{R}$. □

(b) Pick $y_1 \in \mathbb{R}$ and define $y_{n+1} = f(y_n)$. Show that (y_n) is a Cauchy sequence.

Proof. We want to show that $\forall \epsilon > 0, \exists N \in \mathbb{N}$ such that for all $m > n > N$, $|y_m - y_n| < \epsilon$.

For $m > n$, we repeatedly apply the contraction inequality $n - 1$ times:

$$|y_m - y_n| \leq c|y_{m-1} - y_{n-1}| \leq \cdots \leq c^{n-1}|y_{m-n+1} - y_1|$$

Let $k = m - n + 1$. By the triangle inequality:

$$\begin{aligned} |y_k - y_1| &\leq |y_k - y_{k-1}| + |y_{k-1} - y_{k-2}| + \cdots + |y_2 - y_1| \\ &\leq (c^{k-2} + c^{k-3} + \cdots + 1)|y_2 - y_1| \\ &< |y_2 - y_1| \sum_{j=0}^{\infty} c^j = \frac{|y_2 - y_1|}{1 - c} \end{aligned}$$

Setting $M = \frac{|y_2 - y_1|}{1-c} > 0$, we have:

$$|y_m - y_n| < c^{n-1}M$$

Since $0 < c < 1$, $\lim_{n \rightarrow \infty} c^{n-1} = 0$. Given $\epsilon > 0$, choose $N \in \mathbb{N}$ such that $c^{n-1} < \frac{\epsilon}{M}$ for all $n > N$.

Then for all $m > n > N$:

$$|y_m - y_n| < \left(\frac{\epsilon}{M}\right)M = \epsilon$$

Therefore, (y_n) is a Cauchy sequence. □

(c): Prove y is a fixed point of f (i.e., $f(y) = y$) and it is unique in this regard.

Proof. We have $y_{n+1} = f(y_n)$. Set $y = \lim_{n \rightarrow \infty} y_n = \lim_{n \rightarrow \infty} y_{n+1}$, as (y_n) is convergent due to being a Cauchy sequence.

Taking limits on both sides:

$$\lim_{n \rightarrow \infty} y_{n+1} = \lim_{n \rightarrow \infty} f(y_n)$$

Since f is continuous on $\mathbb{R}$:

$$\begin{aligned} \lim_{n \rightarrow \infty} f(y_n) &= f\left(\lim_{n \rightarrow \infty} y_n\right) \\ &\implies y = f(y). \end{aligned}$$

To show that y is unique, let (a_n) and (b_n) be convergent sequences such that $a_{n+1} = f(a_n)$ and $b_{n+1} = f(b_n)$, where $a_n \rightarrow a$ and $b_n \rightarrow b$.

By the Algebraic Limit Theorem:

$$a = b + \lim_{n \rightarrow \infty} (a_n - b_n)$$

Evaluating the limit:

$$\lim_{n \rightarrow \infty} (a_n - b_n) \leq \lim_{n \rightarrow \infty} |a_n - b_n| \leq \lim_{n \rightarrow \infty} c^{n-1}|a_1 - b_1| = 0$$

Thus, $\lim_{n \rightarrow \infty} (a_n - b_n) = 0$.

$$\implies a = b$$

Therefore, y is unique. □